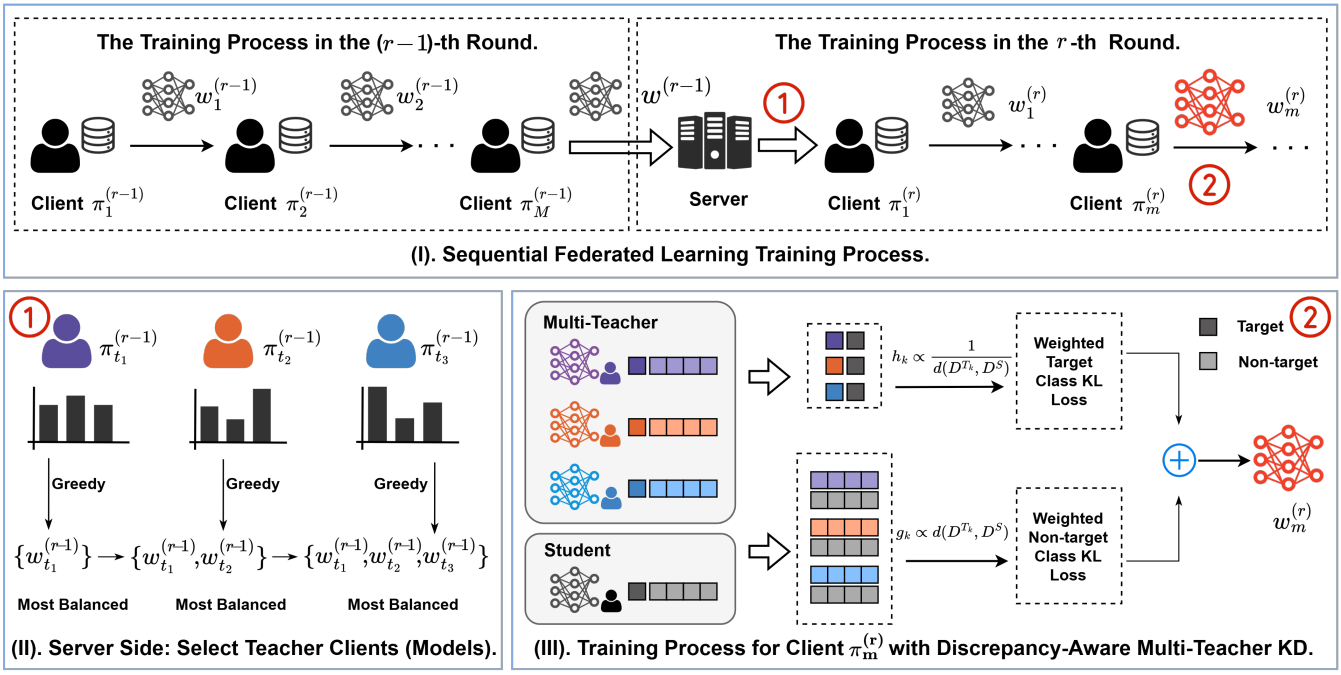


Fig. 2. The overview of our proposed SFedKD. The model is trained in a sequential manner across clients (I). At the beginning of the r-th round, the server applies a greedy teacher selection algorithm to select teacher clients and collect corresponding teacher models from the $(r-1)$-th round (II). The teacher models guide the training process in the r-th round utilizing our proposed discrepancy-aware multi-teacher KD (III).

E. Knowledge Distillation in Federated Learning

Existing techniques applying Knowledge Distillation (KD) to FL can be primarily categorized into two streams. The first employs local models as teachers to guide the global model [51], [52], aiming to integrate knowledge from all clients. However, this typically requires the server to possess a high-quality unlabeled proxy dataset, which is often impractical in real-world scenarios; even subsequent data-free methods [53] suffer from prohibitive computational costs. The second category utilizes the global model as a teacher for local models [32], [54]; however, local data distribution shifts make it difficult for clients to effectively inherit the generalization capability of the global model.

Furthermore, these works predominantly focus on the PFL framework. However, due to the difference in training modes, the KD-based algorithms used in PFL cannot be directly transferred to SFL. Therefore, we propose SFedKD, which introduces multi-teacher KD into the SFL framework. Considering the challenges of uneven teacher contributions and teacher redundancy, our method explicitly considers the complementarity among teacher models and assigns personalized weights. In this way, SFedKD leverages the superior convergence of SFL while effectively overcoming catastrophic forgetting, thereby enhancing the overall convergence of FL.

III. Problem Formulation and Framework

We consider a typical FL system, consisting of a server and a client set $\mathcal{N} = \{1, 2, \cdots, N\}$. Each client $n \in \mathcal{N}$ holds its local dataset $\mathcal{B}_n = \{\xi_i^n\}_{i=1}^{|\mathcal{B}_n|}$ with the size of $|\mathcal{B}_n|$, where ξ_i^n denotes a data sample from $\mathcal{B}_n$. The local class distribution of n-th client's dataset is defined as $\mathbf{D}_n \in \mathbb{R}^C$, where the c-th element $\mathbf{D}_{n,c} = \frac{|\{(\mathbf{x}_i, y_i)|y_i=c\}|}{\mathcal{B}_n}$, $c \in \{1, 2, \cdots, C\}$ is the data proportion of the c-th class. We define $\mathcal{L}(\mathbf{w}; \xi)$ as a loss function to evaluate a model $\mathbf{w}$ which performs on the data sample ξ. The optimization objective is formulated as:

$$\min_{\mathbf{w}} \sum_{n=1}^{N} \mathbb{E}_{\xi \sim \mathcal{B}_n}[\mathcal{L}(\mathbf{w}; \xi)]. \tag{1}$$

In this work, we propose an SFL framework, SFedKD, along with a discrepancy-aware multi-teacher KD approach to address the catastrophic forgetting problem in traditional SFL and a teacher selection mechanism to filter out redundant teacher models for efficient distillation. The overview of our framework is demonstrated in Fig. 2. We consider that the whole SFL process consists of R rounds, and let $r \in \{1, 2, \cdots, R\}$ indicate the rounds. The SFedKD framework works as follows.

At the beginning of each round r, the server randomly samples a sequence of clients from the set $\mathcal{N}$ through sampling without replacement, denote as $\Pi^{(r)} = \{\pi_1^{(r)}, \pi_2^{(r)}, \cdots, \pi_M^{(r)}\}$. Here, $\pi_m^{(r)}$ is the index of the m-th sampled client in the r-th round and M clients are sampled in each round. The m-th trained local model in the sequence during round r is denoted by $\mathbf{w}_m^{(r)}$. The server distributes the global model $\mathbf{w}^{(r-1)}$ obtained at the end of the previous round to the first client in the sequence to start the current round of training by setting $\mathbf{w}_0^{(r)} = \mathbf{w}^{(r-1)}$ (Fig. 2 (I)). Meanwhile, the server will select a subset of teacher clients $\mathcal{T}^{(r)} = \{\pi_{t_1}^{(r-1)}, \pi_{t_2}^{(r-1)}, \cdots, \pi_{t_K}^{(r-1)}\} \subseteq \Pi^{(r-1)}$ from the previous round through the teacher selection mechanism and collect their training models $\mathcal{W}^{(r)} = \{\mathbf{w}_{t_1}^{(r-1)}, \mathbf{w}_{t_2}^{(r-1)}, \cdots, \mathbf{w}_{t_K}^{(r-1)}\}$

TABLE II
TEST ACCURACY (%, MEAN±STD ON 5 TRIALS) COMPARISON OF OUR SFEDKD METHOD TO OTHER BASELINES ON SEVERAL HETEROGENEOUS SETTINGS AND DATASETS. FEDSEQ, CWC AND OUR SFEDKD ARE SFL METHODS, WHILE OTHER BASELINES ARE ALL PFL METHODS. EXPERIMENTS SHOW THAT SFEDKD CONSISTENTLY OUTPERFORMS THESE STATE-OF-THE-ART METHODS.

Method	Fashion-MNIST		CIFAR-10		CINIC-10		CIFAR-100	HAM10000
	Exdir(2,0.5)	Exdir(2,10.0)	Exdir(2,0.5)	Exdir(2,10.0)	Exdir(2,0.5)	Exdir(2,10.0)	Exdir(20,1.0)	Exdir(2,1.0)
FedAvg	76.00±2.37	82.91±0.71	32.48±2.01	54.19±1.27	28.90±1.16	43.93±0.80	33.11±0.22	48.23±1.06
FedProx	76.13±2.16	82.89±0.78	32.48±2.02	53.69±1.25	29.25±1.02	44.00±0.62	33.00±0.13	48.08±0.92
FedCurv	75.73±2.25	82.95±0.83	32.34±1.85	54.15±1.04	28.81±1.18	43.91±0.74	32.99±0.33	47.94±1.06
FedNTD	76.18±3.05	82.62±0.86	32.27±1.40	53.87±0.54	21.21±1.18	38.56±0.83	35.21±0.42	53.56±0.75
FedEXP	73.61±3.06	77.58±0.92	32.59±1.04	52.50±2.18	25.31±0.84	40.41±1.09	27.05±0.94	44.61±0.86
FedDisco	78.22±1.59	82.83±0.67	37.14±1.95	53.56±1.46	31.79±0.93	44.43±0.69	33.65±0.41	50.06±0.85
FedLWS	78.48±0.60	82.41±0.60	29.82±1.20	50.61±1.03	29.53±0.69	43.57±0.39	32.42±0.17	47.20±0.65
FedSeq	81.81±1.37	82.56±0.53	54.66±2.85	65.40±1.09	43.22±0.61	52.49±0.33	33.01±0.70	62.88±5.64
CWC	81.85±1.38	82.94±0.37	55.16±1.52	65.63±1.33	44.61±1.40	52.27±0.58	32.28±0.88	59.06±3.20
SFedKD	**83.66±1.42**	**83.42±1.12**	**61.56±2.32**	**72.77±1.07**	**46.34±0.46**	**55.53±0.42**	**40.15±0.71**	**65.28±0.59**

TABLE III
TEST ACCURACY (%, MEAN±STD ON 5 TRIALS) COMPARISON OF OUR SFEDKD METHOD TO OTHER BASELINES ON SEVERAL HETEROGENEOUS SETTINGS WITH DIFFERENT MODEL ARCHITECTURES ON CIFAR-10.

Method	CNN		ResNet-10		ResNet-18		ResNet-34	
	Exdir(2,0.5)	Exdir(2,10.0)	Exdir(2,0.5)	Exdir(2,10.0)	Exdir(2,0.5)	Exdir(2,10.0)	Exdir(20,1.0)	Exdir(2,1.0)
FedAvg	50.33±0.36	63.97±1.02	32.72±0.47	45.84±1.31	34.15±0.30	48.98±1.57	35.49±0.75	50.42±0.93
FedProx	50.69±0.35	63.90±0.99	33.28±0.55	45.44±1.44	34.61±0.23	48.55±1.50	36.00±0.64	49.84±0.76
FedCurv	46.97±0.43	63.46±0.87	31.83±0.51	45.24±1.66	33.04±0.52	48.28±1.67	34.47±0.72	50.97±1.29
FedNTD	52.61±1.06	66.45±0.84	37.13±1.60	54.22±0.45	38.64±1.38	55.35±1.28	39.64±1.69	57.81±1.75
FedEXP	44.69±1.42	58.03±1.43	28.45±0.72	46.23±1.46	29.83±1.38	49.94±1.92	29.43±1.70	51.39±2.18
FedDisco	53.96±1.32	63.89±0.84	36.37±0.99	46.56±1.58	37.50±1.11	48.47±1.97	39.02±1.35	51.14±1.11
FedLWS	52.91±1.49	63.36±0.95	35.85±0.76	44.55±1.29	36.80±0.96	48.05±1.45	38.31±1.21	49.48±0.96
FedSeq	57.28±1.80	63.52±1.38	42.21±1.18	55.13±1.93	47.40±1.94	64.86±0.31	50.92±1.39	70.12±2.04
CWC	57.10±1.58	63.10±1.41	42.88±1.97	55.61±1.54	47.68±2.30	64.39±2.49	50.83±1.61	72.19±0.89
SFedKD	**62.69±2.08**	**71.38±1.32**	**45.03±1.67**	**64.32±1.57**	**51.60±1.61**	**72.02±0.99**	**53.09±2.15**	**74.63±1.06**

3) Data Partition: To simulate the heterogeneous data distribution, we use an extended Dirichlet strategy [19] to partition the datasets. The extended Dirichlet strategy can be denoted by ExDir(C, α), which introduces an additional parameter C to determine the number of classes per client compared with the Dirichlet distribution [33]. Specifically, we randomly allocate C different classes to each client and then draw $\mathbf{p}_c \sim \text{Dir}(\alpha \mathbf{q}_c)$ to allocate a proportion of the samples of class c to each client. For example, $\mathbf{q}_c = [1, 1, 0, 0, \ldots,]$ means that the samples of class c are only allocated to the first 2 clients. This strategy significantly enhances the flexibility of our design for federated learning in heterogeneous scenarios. A visualization of the data partitions for the CIFAR-10 at varying settings can be found in Fig. 4. For the Fashion-MNIST, CIFAR-10, and CINIC-10 datasets, we configure two settings: ExDir(2,0.5) and ExDir(2,10.0), while for CIFAR-100 and HAM10000, we implement ExDir(20,1.0) and ExDir(2,1.0), respectively. As the HAM10000 dataset has a lower class number than other datasets, we only set one case for it, and so does CIFAR-100 due to its large number of classes.

4) Implementation Details: All the experiments are implemented with Pytorch and conducted on a system with an Intel(R) Xeon(R) Platinum 8352V CPU, an NVIDIA A40 (48GB) GPU, and 256GB RAM. In our setting, we divide the dataset into 100 clients, then randomly sample 10 clients and select 5 teachers in each round. We run federated learning for 1000 rounds. The number of local epochs and batch size are 5 and 64, respectively. We use SGD optimizer with a 0.01 learning rate and the weight decay is set to 1e-4. To calculate the distance between data distributions, we employ KL-Divergence to measure the discrepancy. Before computing the divergence, the input distributions are normalized, and zero entries are replaced with a small value of 1e-10 and then renormalized to avoid numerical instability caused by log(0).

5) Baselines: We compare SFedKD with two SFL baselines: **FedSeq** [19], which is the most representative SFL algorithm; **CWC** [27], a method based on cyclical weight consolidation to mitigate catastrophic forgetting in SFL. In addition, we compare our method with seven PFL baselines: **FedAvg** [1], which is the pioneering PFL method; **FedProx** [30], a classic regularization-based PFL method; **FedCurv** [31]: a PFL method based on curvature adjustment regularization; **FedNTD** [32], which preserves the global knowledge by not-true distillation in PFL; **FedEXP** [34], which speeds up PFL convergence via extrapolation; **FedDisco** [33], whose aggregation weights involve both the dataset size and the